Environmental Satellite Processing Center (ESPC)



NDE to AWIPS
Interface Control Document (ICD)

NPP Data Exploitation (NDE)
Advanced Weather Interactive Processing System (AWIPS)

Version 1.0

September 27, 2012

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Version 1.0

September, 2012

Approval Pages

Document Title Block:

Environmental Satellite Processing Center (ESPC)

NPP Data Exploitation (NDE)

NDE to AWIPS Interface Control Document (ICD)

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DOCUMENT HISTORY

DOCUMENT REVISION LOG

The Document Revision Log identifies the series of revisions to this document since the baseline release. This page will become a permanent part of this document.

Version Number	Date	Description of Change/Revision	Section/Pages Affected	Changes Made by Name/Title/Organization
1.0	September 27, 2012	Initial Release	All	Kevin Berberich
1.0	October 3, 2012	Quality Assurance edit	All	Janice Olfus-Carter

VERSION NUMBER IDENTIFIER

The document version number, which also corresponds to the Document Control Number (DCN), identifies whether the document is a working copy, final, revision, or update, defined as follows:

- **Working copy or Draft:** a document not yet finalized or ready for distribution; sometimes called a draft. Use 0.1A, 0.1B, etc. for unpublished documents.
- **Final:** the first definitive edition of the document. The final is always identified as Version 1.0.
- **Revision:** an edition with minor changes from the previous edition, defined as changes affecting less than one-third of the pages in the document. The version numbers for revisions 1.1 through 1.9, 2.1 through 2.9, and so forth. After nine revisions, any other changes to the document are considered an update. A revision in draft, i.e. before being re-baselined should be numbered as 1.1A, 1.1B, etc.
- **Update:** an edition with major changes from the previous edition, defined as changes affecting more than one-third of the pages in the document. The version number for an update is always a whole number (Version 2.0, 3.0, 4.0, and so forth).

Abstract

The NOAA NPP Data Exploitation (NDE) Project is responsible for managing the requirements, design, development, test, and transition to operations of the near real time product generation of NPP satellite data for dissemination to the civilian community. The Advanced Weather Interactive Processing System (AWIPS) is the primary system used by NWS forecasters for the preparation and dissemination of forecasts, watches, warnings, and advisories. An interface is being established between NDE and AWIPS to enable NWS forecaster's access to near real time NPP satellite products. This NDE to AWIPS ICD describes the interfaces and data flows of science data products from the NDE system to NWS' AWIPS system. This document provides interface specifications between the NOAA NDE and the AWIPS to ensure proper sizing of data production facilities, network sizing, file transfer protocol, and software interfaces. This document is also being prepared to facilitate long-term maintenance and evolution of this important NOAA interface.

Further editions of this document will include updated interface functions and responsibilities which will be determined and become more evident as NOAA works through the transition to operations process for the NDE system. When continuous data activation occurs between the NDE to AWIPS system, targeted for early October 2012, NDE will reside under the NESDIS, Office of Systems Development (OSD); however, a transition over to the NESDIS Office of Satellite and Product Operations (OSPO) is targeted to occur during FY2013. When this transition occurs, roles and responsibilities will change, as will some of the verbiage within this document.

Document Overview

The first section of this document sets the context for the ICD in relation to the whole NDE and AWIPS environments. In the second section, reference documents are divided into parent documents from which this document follows or is derived, applicable documents for additional information relevant to this document, and informational documents for details outside the scope of this ICD. Relevant interface specifics such as the network physical and logical topologies are contained in the third section with communication transfer mechanisms, specific message and file contents, timeliness, and recovery mechanisms in sections 4 and 5. NDE Data Specifications such as data volume and file naming conventions are described in section 6.

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1. Introduction

1.1 Scope

This document provides detailed specifications for the interface between the National Environmental Satellite, Data, and Information Service (NESDIS) Environmental Satellite Processing Center (ESPC) NPP Data Exploitation (NDE) subsystem and the National Weather Service (NWS) Advanced Weather Interactive Processing System (AWIPS) Network Control Facilities (NCF). This document will include descriptive details of any physical and logical interfaces along with any environmental satellite products and information that are transferred across them.

This document will be updated, as required, to describe additional products (as they become available) and any changes to the system interfaces.

1.2 Background

The current legacy series of National Oceanic and Atmospheric Administration (NOAA) Polar-orbiting Operational Environmental Satellite (POES) will be replaced by the next generation series of environmental monitoring satellites acquired through the Joint Polar Satellite System (JPSS) Program Office. Suomi National Polar-orbiting Partnership (S-NPP), a NASA satellite, is in some respects a bridge mission from NOAA's current series of polar-orbiting satellites to JPSS. Although S-NPP is designated a Risk Reduction mission, NOAA is "operationalizing" the products derived from S-NPP sensors to ensure continuity of the POES mission. After NPP, the JPSS Program Office will acquire two additional satellites, called JPSS-1 and JPSS-2 that will have sensor capabilities similar to those found on NPP.

The NDE subsystem of ESPC is the only source of near-real time global environmental products derived from S-NPP sensors dedicated to civilian applications, and is the only source of S-NPP products tailored for NOAA's requirements, that is NOAA Unique Products (NUPs) and Visible Infrared Imaging Radiometer Suite (VIIRS) imagery products. Alternative sources of near-real time S-NPP data include Direct Readout ground stations, which provide regional products, and systems operated by the US Department of Defense dedicated to military applications. Retrospective S-NPP data (including selected NUPs from NDE) are available from NOAA's Comprehensive Large Array-data Stewardship System (CLASS) no sooner than six hours after observation.

The AWIPS is the primary system used by NWS forecasters for the preparation and dissemination of forecasts, watches, warnings, and advisories. Such NWS forecast products are based in part on various satellite products provided by NESDIS satellites and ground systems (such as S-NPP and ESPC/NDE). The NWS AWIPS NCF is the primary AWIPS subsystem that receives environmental satellite products from systems such as ESPC/NDE. A new NDE-to-AWIPS interface is being established to enable the transfer of S-NPP products to AWIPS. This ICD has been prepared to document the interface, to facilitate its implementation and long-term maintenance, and to communicate information about the products that NDE provides to AWIPS across this interface.

1.3 Context - Physical

S-NPP and JPSS satellites will provide global-coverage satellite data to four central processing facilities, called "Centrals." NOAA's NESDIS operates the NESDIS Central. The Air Force Weather Agency (AFWA), Fleet Numerical Meteorology and Oceanography Center (FNMOC), and Naval Oceanographic Office (NAVOCEANO) operate the other three Centrals. The NESDIS Central is hosted at the NOAA Satellite Operations Facility (NSOF) in Suitland, Maryland, and consists of the collocated JPSS Integrated Data Processor (IDP), the JPSS Mission Management Center (MMC), and the NDE system. The NOAA NDE is hosted as a subsystem within the NOAA ESPC.

The following diagram, from the NDE Concept of Operations, emphasizes the NESDIS side of the NDE/AWIPS interface. The only flows relevant to this ICD are labeled "VIIRS Imagery Products" in Figure 1-1. The remaining flows are described in the above referenced operation document.

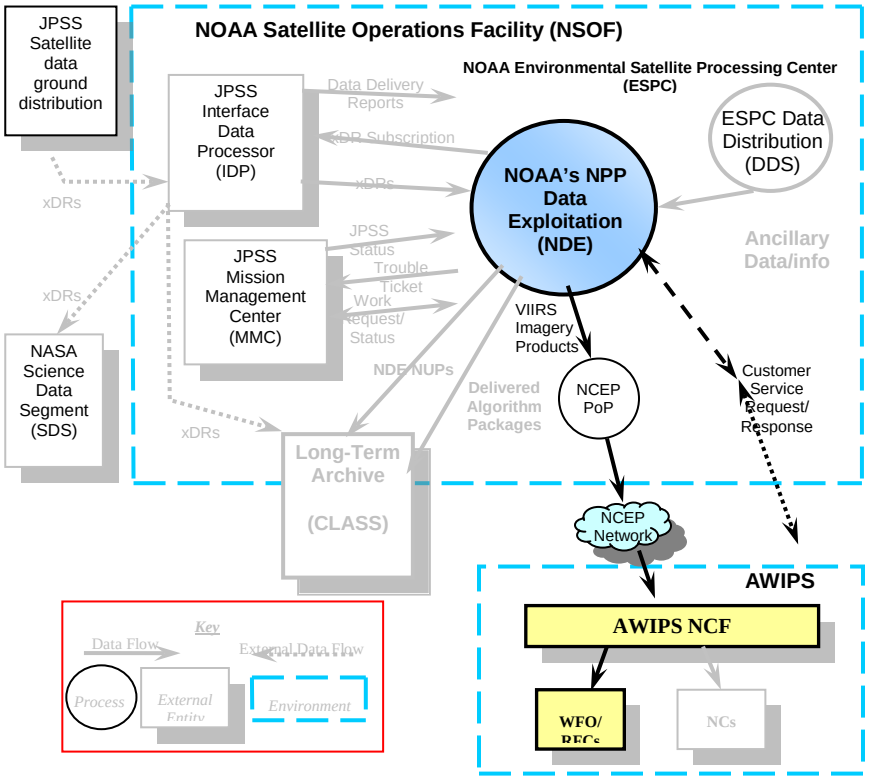


Figure 1-1: High Level View of the NDE/AWIPS Interface

AWIPS is the primary system used for the preparation of NOAA's weather warnings, forecasts, watches, advisories and other hydrometeorological products. NOAA's National Weather Service operates the AWIPS system, which receives environmental satellite products from NESDIS.

AWIPS also receives various products from other sources. The AWIPS NCF is the main entry point for products flowing into AWIPS. Thus, the AWIPS side of the NDE/AWIPS interface primarily concerns the NCF, and the networks connecting NDE to the AWIPS NCF.

Downstream from the NCF, satellite and other products are disseminated to numerous NWS/AWIPS field nodes, where forecasts and other NWS products are generated. These products, in turn, are provided to internal and external users via the AWIPS communications network and other NOAA dissemination systems. Figure 1-2 provides a greatly simplified AWIPS context depiction. Section 1.7, below, provides more-detailed information about AWIPS.

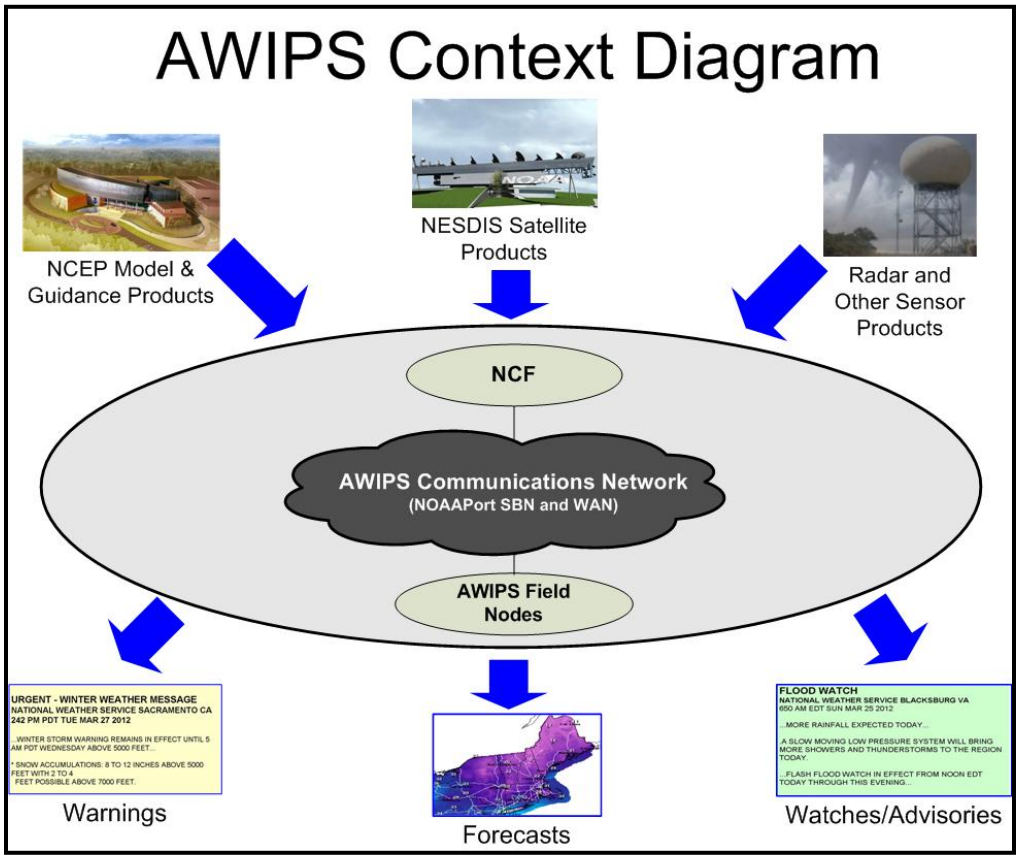


Figure 1-2: AWIPS Context Diagram

1.4 NDE Overview

The NDE will ingest S-NPP sensor, temperature, environmental, intermediate and application related data records, collectively referred to as xDRs, from the JPSS IDP, the instance of the IDPS system at the NSOF. NDE will use xDR information and generate a set of VIIRS satellite imagery products. NDE will tailor and distribute these VIIRS imagery products according to the production rules and subscriptions defined in its database. The NDE database will contain the definitions of all products, algorithms, product generation rules, files, user profiles, subscriptions, interface configurations and resources used by the NDE Data Handling System (DHS). Interactive web-based user interfaces will allow authorized NDE users to review their orders; however, data requests will go through existing OSPO processes. The NDE DHS will generate products in response to the arrival of data from the IDP (e.g. queue an algorithm or a distribution process) as defined by the ingest rules, production rules, and subscriptions defined in the NDE database.

The NDE DHS will distribute products based upon the rules specified in the users' subscriptions.

The NDE DHS subscription database will contain a list of xDRs, NUPs and VIIRS Imagery products along with tailoring options created by the NDE. The subscription may also contain optional parameters that limit the distribution of product by spatial coverage area and data quality thresholds. The subscription parameters may also specify the type of compression of the delivered file and delivery methods, and notification options.

1.5 NDE System Mission Objectives

NOAA's NDE system will receive data from the JPSS IDP, process and package it to meet user requirements, ensure appropriate VIIRS imagery products are archived, distributed, or provided for distribution of data to authorized users, and provide online services to users of its products. NDE's primary mission is to provide products derived from S-NPP observations to NOAA's operational and climate communities and to other civilian users. In order to fulfill the above mission, NDE will acquire the resources necessary to achieve the following objectives:

- Disseminate S-NPP Data Records to users delivered by the JPSS program office.
- Generate and disseminate tailored S-NPP Data Records (versions of S-NPP Data Records in previously agreed alternative formats and views).
- Generate and disseminate VIIRS imagery products (augmented environmental products constructed from S-NPP Data Records).
- Deliver NUPs and VIIRS imagery products, product processing elements, and associated metadata to the Comprehensive Large Array-data Stewardship System (CLASS) for long-term archiving.
- Provide services to users, including NDE product training, product enhancement, and implementation support across NOAA.
- Provide software and tools for S-NPP Data Record and VIIRS imagery Product reformatting and other data manipulations.

In order to support these objectives, the NDE DHS is partitioned into Production, System Test, and Development Environments. The NDE Project will use the System Test environment to test the interface with AWIPS. The NDE Project will use the Production environment to deliver VIIRS imagery products and tailored products to AWIPS on an operational basis; however, the initial delivery of test products from NDE will be provided to AWIPS via NDE's System Test Environment which is the contingency plan for not having its Production Environment fully installed, configured and adequately integrated within ESPC in time for initial data delivery efforts. When the Production Environment is deemed ready for operational use, NDE will transition data delivery functions from its System Test Environment to the Production Environment.

NDE VIIRS imagery products and tailored products will be declared operational by the NESDIS Satellite Products and Services Review Board (SPSRB). When a NUP is declared operational, a set of four SPSRB Standards and Guidelines documents intended for its' operational use will be made available to users. These documents include the External Users Manual, Internal Users Manual, System Maintenance Manual, and Algorithm Theoretical Basis Document (ATBD).

1.6 NDE/AWIPS Connection Overview

There are networks used to move S-NPP products from NDE to AWIPS which are intrinsic to neither the NDE nor the AWIPS system. These networks include the COPC/NCEP Wide Area Network (WAN) and OPSnet. For the AWIPS ANCF, the NESDIS-to-NWS handoff of NDE products occurs at the NCEP network point of presence (PoP) at the NSOF facility. There is a primary network path used to move NDE products to AWIPS ANCF. Once connectivity and functionality at the Backup NCF (BNCF) is implemented, in general, only one product stream will be active at any given time (i.e., the primary or the backup, but not both). For the majority of the time, the primary network pathway will be used. For ANCF, the primary network pathway extends from the NCEP PoP (at NSOF) through an intermediate contractor facility, to the NWS Telecommunications Operations Center (TOC) in SSMC-2/NWS (Silver Spring, MD). The NWS OPSnet network will provide the connectivity between the TOC and the AWIPS NCF.

The secondary network pathway is similar to the primary path, except that the network extends from NCEP PoP at NSOF to Fairmont WV, where the AWIPS BNCF is located. An OPSnet PoP at Fairmont will provide the necessary connectivity between the COPC/NCEP WAN and the BNCF. This network path will not be made available for the NDE to AWIPS backup data flows in the short term; however, future plans call for this backup scenario.

1.7 AWIPS Overview

AWIPS is a central component of the NWS warning and forecasting services operations. It provides an integrated suite of powerful automated data processing equipment and state-of-the-art WAN and satellite communications services, and provides the user interface and underlying IT infrastructure that NWS forecasters use to prepare and disseminate most key NWS weather products. AWIPS has been deployed to NWS sites throughout the continental United States and in Alaska, Hawaii, Puerto Rico, and Guam. AWIPS is a highly distributed system used for the collection, generation, and dissemination of numerous hydrometeorological data and products throughout North America.

The NWS side of the NDE/AWIPS interface is now briefly described. As indicated in preceding subsections, the AWIPS NCF is the primary NWS subsystem that receives environmental satellite products from systems such as ESPC/NDE. The NCF provides telecommunications services, performs data archiving, quality control, backup and replenishment of data sources for the NWS field sites, and monitoring. In addition, the NCF and its staff identify, diagnose, and correct system faults and outages; monitor site equipment operations; and alert site users when malfunctions or degradations occur. The NCF maintains a time source and regularly coordinates network-level synchronization. In support of system administration, the NCF serves as the focal point for the troubleshooting of all software baseline and maintenance release installations at the field sites.

The AWIPS NCF actually includes three separate instances: the ANCF, the BNCF and the Test NCF (TNCF). The ANCF is located in Silver Spring Metro Center-2 (SSMC-2), which is also NWS Headquarters, in Silver Spring, MD. The BNCF is located at a National Aeronautics and

Space Administration (NASA) Independent Verification and Validation (IV&V) Facility in Fairmont, WV. The TNCF is located at the AWIPS contractor facility, adjacent to NWS/SSMC-2.

The NCF receives numerous product streams, for relay to AWIPS systems at NWS field offices. In addition to receiving S-NPP products from NDE, the NCF receives numerous other products (e.g., model products from the National Centers for Environmental Prediction, NCEP). The NCF receives products and prepares the data stream for dissemination to NWS field sites and external users. The data stream is sent downstream from the NCF to a contractor Master Ground Station (MGS) for uplink onto the NOAAPort Satellite Broadcast Network (SBN). The SBN is a vendor-provided NOAA product broadcast servicing much of the Northwest hemisphere quadrant. This service is primarily intended to reach NWS field offices, however a broad NOAA-external user community also receives and utilizes this data stream (via their own equipment and software). Most NWS field offices have a satellite downlink antenna, which brings the product stream into one or more AWIPS systems. A simplified representation of this data stream appears in Figure 1-3. As alluded to above, once implemented, the secondary network pathway will be occasionally used to provide S-NPP products to the BNCF and the Backup MGS (BMGS) in Fairmont, WV. In backup scenarios, the BNCF and BMGS provide the data stream to the NOAAPort SBN.

As described above, the main AWIPS subsystems most pertinent to the NDE/AWIPS interface are the ANCF, BNCF, and TNCF. AWIPS also includes over 150 field node systems, including 122 Weather Forecast Office (WFO) configurations, 13 River Forecast Center (RFC) configurations, 7 NCEP configurations, 6 Regional Headquarters systems, 4 training systems, and developmental/test systems. There are two major components to the AWIPS communications network. The first component is the NOAAPort SBN (mentioned above). The second AWIPS communications component is the AWIPS WAN, which is a (mostly terrestrial) MPLS WAN that interconnects all field nodes and the NCF systems. The AWIPS WAN is managed as part of OPSnet.

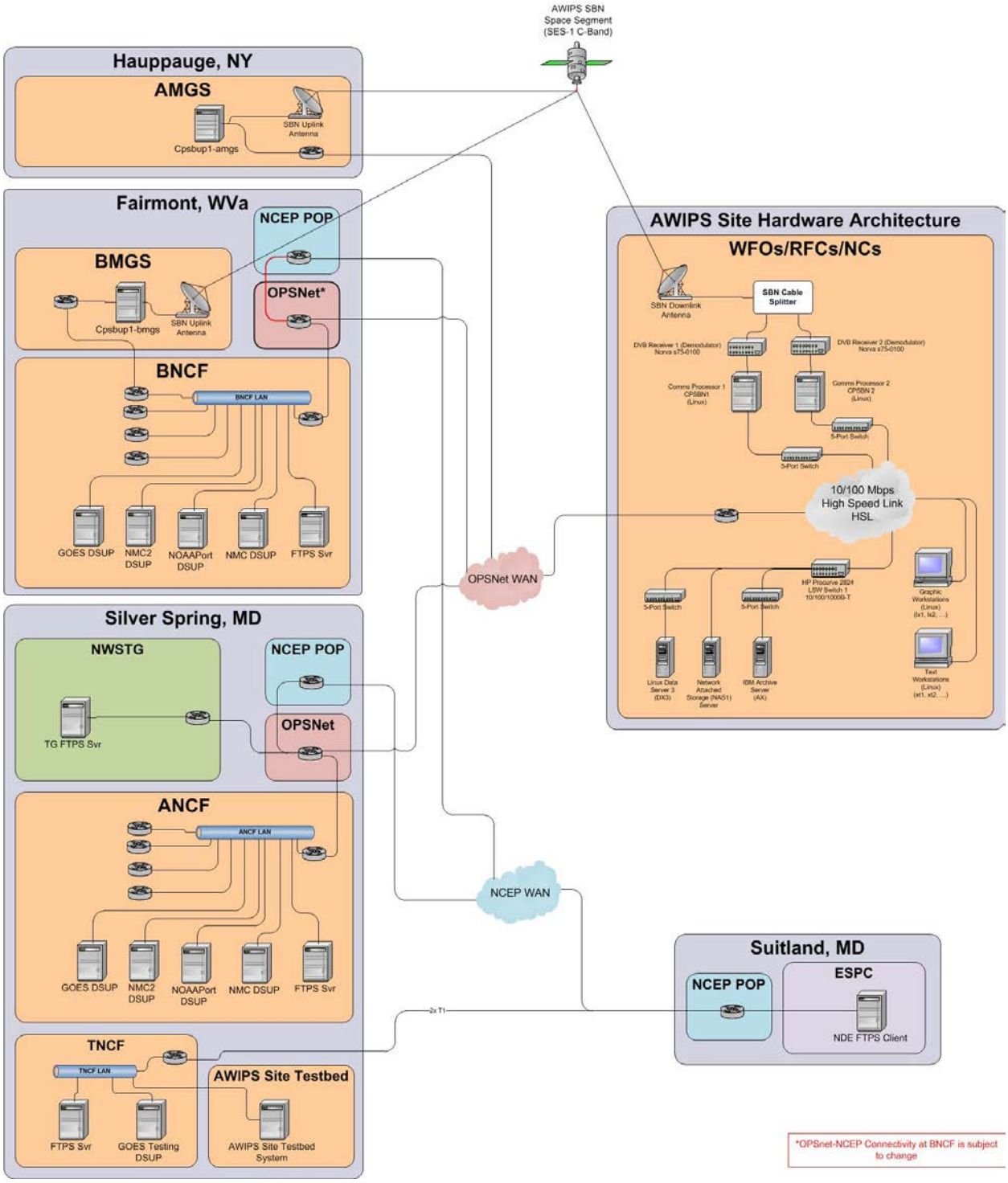


Figure 1-3: Data Flow Into AWIPS Systems

1.8 Configuration Management

Version 1.0 and subsequent versions of this document are maintained under NDE configuration management (CM), ESPC CM and AWIPS CM. Proposed changes to the body of the document will be presented for review and approval to the NDE CCB, the AWIPS CCB, ESPC CCB and

appropriate Archive personnel as shown on the Approval page. The NDE CCB, in coordination with ESPC CCB and AWIPS CCB and Archive personnel, rules on the final acceptance of proposed document changes. The CCB contact information for the three organizations is provided in Appendix B.

2. References

2.1 Parent Documents

These documents are available from the NSOF library or NDE configuration management.

Document Source and Number	Document Title
OSDPD/ESPC-NDE-008-018-005-V1.0	NDE System Requirements Specification
OSDPD/ESPC-NDE-008-024-015-V4.0	NDE Concept of Operations

Table 2-1: List of Parent Documents

2.2 Applicable Documents

These documents are available from the NSOF library or NDE configuration management.

Document Source and Number	Document Title
OSDPD/ESPC-NDE-008-016-003-V2.0	NPOESS Preparatory Project (NPP) Data Exploitation (NDE) Configuration Management Plan
ESPC-CM1301-05-CT-0044 (v1.3)	ESPC Configuration Management Plan and Procedures
(TBA) v4.3	ESPC (NOAA5045) System Security Plan
http://www.ospo.noaa.gov/Organization/About/access.html	NESDIS Policy on Access and Distribution of Environmental Satellite Data and Products
v1.0	Interconnection Security Agreement Between NESDIS OSPO ESPC (NOAA5045) and NWS OST AWIPS (NOAA8107)
https://access.portal.noaa.gov/doc/CIO/ITSITnew/CITR_009_Password_Requirements.pdf	Department of Commerce Interim Technical Requirements (CITR) CITR-009: Password Requirements Version: 1.1
http://www.ietf.org/rfc/rfc4217.txt RFC 4217	Securing FTP with TLS
http://projects.osd.noaa.gov/spsrb/standards_docs/SPSRB_NetCDF_Policy.pdf	NOAA/NESDIS SPSRB Policy on NetCDF
ISO 8601:2004	Representation of dates and times
NDE_template_v1_official.ncml	The STAR-NDE Granule Level Metadata Template
v1.0	NWS Agreement to Deliver NPP Data to AWIPS and the NWSTG via the COPC Network

Table 2-2: List Of Applicable Documents

2.3 Informational Documents

Documents are available from Internet sources, where noted, or through NDE configuration management.

Document Source and Number	Document Title
ESPDS-DAP-WA2-DOC-2.0	Standards for Algorithm Delivery and Integration Using Delivered Algorithm Packages (DAPs)
http://www.fgdc.gov/metadata/csdgm/ FGDC-STD-001-1998	Content Standard for Digital Geospatial Metadata
http://www.fgdc.gov/standards/projects/FGDC-standards-projects/csdgm_rs_ex/remote-sensing-metadata FGDC-STD-012-2002	Content Standard for Digital Geospatial Metadata: Extensions for Remote Sensing Metadata
ISO 19115	Geographic information – Metadata
https://geo-ide.noaa.gov/wiki/index.php?title=Formats_for_delivery_of_scientific/environmental_data	Formats for Delivery of Scientific/Environmental Data
http://projects.osd.noaa.gov/spsrb/doc_standards_guildlines.htm	NOAA/NESDIS Satellite Products and Services Review Board (SPSRB) Documentation Standard Guidelines

Table 2-3: List of Informational Documents

3. Network Connectivity

3.1 Context – Operational Summary

Entities of concern for this NDE-AWIPS ICD are the NDE DHS as shown in bold outline in Figure 3-1, AWIPS Network Control Facility (NCF), the AWIPS users of the S-NPP data products, the Weather Forecast Offices (WFOs), the River Forecast Centers (RFCs), and the NCEP National Centers (NC). The NDE DHS continuously generates NUPs and tailored products in a near real-time manner and delivers these products via digital communications networks to the AWIPS. The NCF forwards these data products via either the SBN broadcast network or its MPLS terrestrial network to its WFO, RFC and NC users.

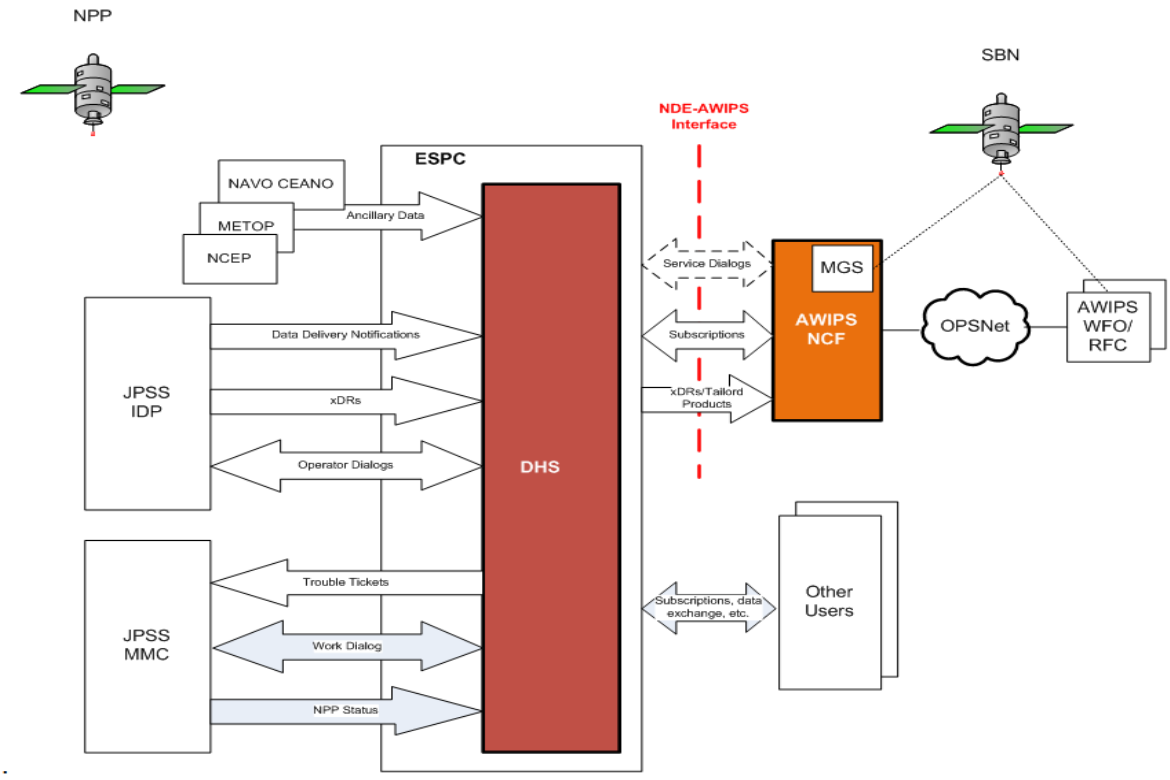


Figure 3-1 Flow Diagram

3.2 Physical Environment

This section provides an overview of the network connectivity between NDE and AWIPS systems. As documented in the *NWS Agreement to deliver NPP Data to AWIPS and the NWSTG via the NCEP/COPC Network* (Date: March 19, 2012), the NCEP/COPC network will be used to disseminate NPP data into the NWS Telecommunication Operations Center (TOC) for relay to AWIPS and other NWS systems. NDE servers will forward (i.e., execute a push of the data) S-NPP data products to AWIPS FTPS servers located at the ANCF in Silver Spring, Maryland (MD), BNCF in Fairmont, West Virginia (WV) and the TNCF at the Raytheon facilities in Silver Spring, MD. At any given time, only one of the ANCF/BNCF instances is actively forwarding

to the SBN. The network that interconnects these sites with each other is managed and maintained by NWS NCEP, OPSnet Team and AWIPS NCF. A number of factors motivated NDE and AWIPS to use these networks: the COPC/NCEP network was already in place, was already present at most physical sites where connectivity between NDE and AWIPS was required, and it provides a large amount of available bandwidth.

The NDE systems located at the NESDIS' ESPC facility at the NSOF building have established a LAN side connection with the NCEP PoP at NSOF. The NCEP PoP provides the point of entry onto the NCEP/COPC network. This connectivity is documented in Figure 3-2 on the following page, but due to its complexity, the details of the NCEP network are not illustrated in this diagram or documented anywhere in this ICD.

At the other end of this communications path, in Silver Spring's TOC data center, NCEP terminates its network with another NCEP router that is connected to the OPSnet networking equipment which provides connectivity to both the AWIPS ANCF and the NWS Telecommunication Gateway (NWSTG). OPSnet is the NWS operational network developed and operated by the TOC. As an intermediary for this connection, the OPSnet network provides network security via a firewall and provides connectivity management as an OPSnet service for both the AWIPS ANCF and the NWSTG.

The current end-to-end network connectivity between NDE distribution systems at the ESPC and the ANCF provide a bandwidth of 100 Mbps. The COPC/NCEP WAN provides a bandwidth of 10 Gbps, the NCEP LAN connection to OPSnet is 100 Mbps and the AWIPS LAN connection to OPSnet is 100Mbps. The entire end-to-end network will be upgraded over the next year is expected to be 1Gbps. Both OPSnet and the ANCF network architectures provide router diversity in order to provide protection against router failure. Currently there is only one COPC/NCEP router at the NWSTG that provides network connectivity from the NDE distribution system. Other network solutions for dynamic and diverse communications are being reviewed at this time. The ANCF routers connect to the ANCF FTPS servers via a LAN.

The ANCF supports S-NPP data distribution with two identical FTPS servers that share a floating IP address. The failover from a primary FTPS server to its backup server is transparent to NDE except that the failover process takes 60 seconds. For this period of time, NDE may lose connectivity to the FTPS servers' floating IP address.

The BNCF is located at a NASA IV&V Facility in Fairmont, WV. Network connectivity to this site will come by way of a data stream flowing over the COPC/NCEP WAN which is already at the facility servicing the NCEP backup Supercomputer. The AWIPS BNCF and OPSnet CE routers are currently working, and it is anticipated that NCEP will establish a LAN side connection to the OPSnet CE Routers. The estimated time frame for completion of this is the first quarter, FY13. This connection will enable NDE to distribute the S-NPP data across the COPC/NCEP WAN to the OPSnet CE routers and then to the AWIPS routers for delivery to the backup AWIPS system. This ICD will be updated once the schedule and technical details for this connection become final.

In addition to the ANCF and BNCF, the AWIPS TNCF is located at the contractor run facility

(currently across from the street from the NWS HQ). Connectivity is established through the AWIPS Wireless Local Area Network (WLAN). Current bandwidth of this connection is about 30Mbps. As depicted in figure 3-2 below, ANCF receives the S-NPP data products through its OPSnet connection then forwards these products to TNCF via a wireless connection.

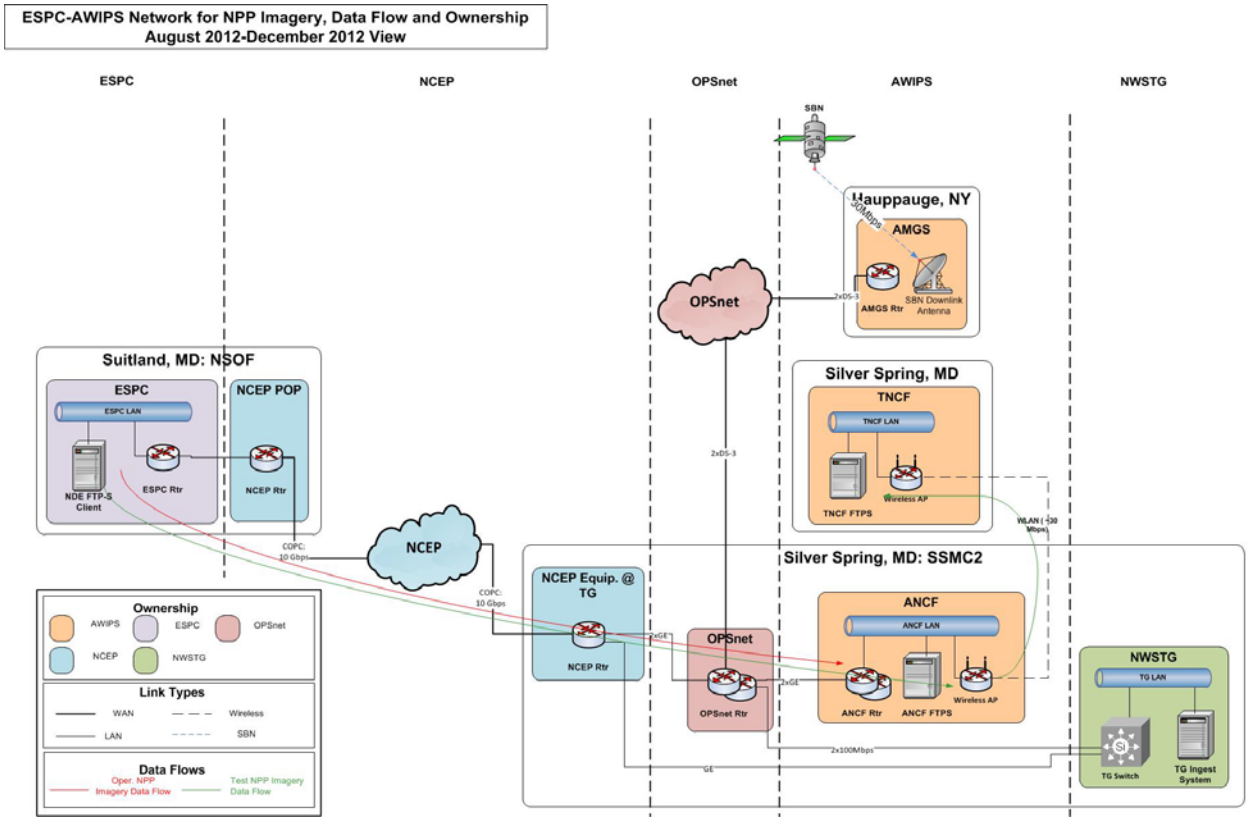


Figure 3-2 Physical Connections

3.3 Demarcation, Roles and Responsibilities

The physical environment that supports the NDE data exchange with AWIPS is owned and operated by five different organizations at NOAA. Figure 3-2 illustrates the ownership of the network by the NOAA stakeholders. The ESPC owns the connected network equipment at NSOF and will take ownership of the NDE FTPS client systems at NSOF once they transition into the operational environment.

Within the scope of this project, ESPC will be responsible for the following:

- Configuring ESPC network equipment in order to allow S-NPP data products to be sent to AWIPS ANCF, BNCF and TNCF
- Providing sufficient bandwidth to the NCEP POP at NSOF such that all agreed to S-NPP data products can be sent to the AWIPS ANCF, BNCF and TNCF

- Working with ANCF, NCEP, OPSnet and AWIPS networking staff to resolve any network problems
- Communicating network scheduled maintenance and system outages with NCEP, OPSnet and AWIPS staff

NCEP's ownership and responsibility begins at its router's ESPC facing network interface at NSOF. NCEP owns and operates the network starting with its network equipment at NSOF, across its NCEP network and through its network equipment at Silver Spring's TOC data center. Their network ownership and responsibility ends at their OPSnet facing network interface.

Within the scope of this project, NCEP will be responsible for the following:

- Configuring NCEP network equipment in order to allow S-NPP data products to be sent to the AWIPS ANCF, BNCF and TNCF
- Given sufficient operation and maintenance (O&M) funding levels continue, NCEP will have the ability to provide 10 Gbps between the NCEP PoP at NSOF and its NCEP equipment in Silver Spring, Maryland.
- Working with ESPC, OPSnet and AWIPS networking staff to resolve any network problems
- Communicating network scheduled maintenance and system outages with ESPC, OPSnet and AWIPS staff

OPSnet's ownership and responsibility begins at the OPSnet CE routers that connect to the NCEP router within the TOC data center located on the 6th floor of SSMC2 at the NWS HQ. The OPSnet team manages and maintains the OPSnet network which provides connectivity to the AWIPS ANCF and BNCF networks. Their network ownership and responsibility is the OPSnet Routers and Firewalls that enable secure connectivity to both AWIPS networks.

Within the scope of this project, OPSnet will be responsible for the following:

- Configuring network equipment in order to allow S-NPP data products to be sent to the AWIPS ANCF, BNCF and TNCF
- Providing sufficient bandwidth such that all agreed to S-NPP data products can be sent to the AWIPS ANCF, BNCF and TNCF
- Working with NESDIS (ESPC & NDE), NCEP, and AWIPS networking staff to resolve any network problems

- Communicating network scheduled maintenance and system outages to ESPC, NCEP and AWIPS staff

3.4 Network Protocols and Standards

NDE and AWIPS will use standard communication protocols in order to distribute S-NPP data products to AWIPS. The following table lists the protocols used.

Interface	Parameter	Value
ESPC(NDE)-ANCF/BNCF/TNCF	Data Link Protocol	Ethernet
	Routing Protocol	N/A (static routes)
	Multicast Support	None
	IP versions supported	IPv4
	Interface Maximum Transmission Unit (MTU)	1500 bytes
	Network Security	FTPS using X.509 Certificates

Table 3-1: OSI Model Specifications

3.5 Network Management Transactions

TBD

3.6 Security Considerations

The entire data path from NDE to AWIPS is internal to NOAA. The networks and circuits are all NOAA assets and protected to the level deemed appropriate for each piece or part of the NOAA infrastructure. For further information pertaining to network security reference the "Interconnection Security Agreement between NESDIS OSPO ESPC (NOAA5045) and NWS OST AWIPS (NOAA8107)".

NDE and AWIPS will use the FTPS protocol for file exchange. The "S" extension of FTP refers to the added security associated with the encryption of the control channel for the exchange of username and password. The data channel and its data are not encrypted.

Certificates are required for security keys used by the FTPS protocol to encrypt the control channel. NESDIS and NWS will create or obtain these certificates through manual exchange between these organizations (i.e., "out of band").

AWIPS will create and assign a username and password for NDE to use for authenticating its FTPS clients at AWIPS' FTPS servers. These credentials will be created to be consistent with the Department of Commerce's security policies and will be stored in the AWIPS security database system.

4. End to End Data Propagation

This section will show the sequence of events from start to finish for the request for a data product from NDE, its preparation, its dissemination, the path of its dissemination through the network and systems, its receipt, its ingest and decoding and display by the AWIPS II system.

4.1 Observation Data from Satellite through ESPC Processing

The Suomi National Polar-orbiting Partnership (S-NPP) satellite carries a diverse payload of scientific instruments to monitor the planet. The 4,600-pound (2,100 kilogram) spacecraft crosses the equator each afternoon at about 1:30 p.m. local time. The S-NPP spacecraft orbits the Earth at an altitude of 512 miles (824 km) and completes over fourteen orbits per day (~102 minutes per orbit), observing the entire earth's surface.

The S-NPP spacecraft carries five key instruments: the Advanced Technology Microwave Sounder (ATMS), the Cross-track Infrared Sounder (CrIS), the Ozone Mapping and Profiler Suite (OMPS), the Visible Infrared Imaging Radiometer Suite (VIIRS), and Clouds and the Earth's Radiant Energy System (CERES). Scientists use ATMS, a 22-channel passive microwave radiometer, to create global models of temperature and moisture profiles that meteorologists enter into weather forecasting models. CrIS, a Michelson interferometer, monitors characteristics of the atmosphere, such as moisture and pressure that are used to produce improvements in both short-and-long term weather forecasting. OMPS, a suite of hyperspectral-imaging spectrometers, measures Earth's ozone levels, particularly near the poles where ozone levels fluctuate the most. VIIRS, a 22-band radiometer, collects visible and infrared views of Earth's dynamic surface processes, such as wildfires, land changes, and ice movement. VIIRS also measures atmospheric and oceanic properties, including clouds and sea surface temperature. Finally, CERES, a 3-channel radiometer measuring reflected solar radiation, emitted terrestrial radiation, and total radiation, monitors the natural and anthropogenic effects on the Earth's total thermal radiation budget.

The S-NPP spacecraft bus contains all of the supporting subsystems needed to control the spacecraft and transmit the collected environmental data to the ground. Representative supporting subsystems include structures, thermal control, power, attitude control, payload data processing and storage, command and control, and communications. The sensors onboard make observations of environmental phenomena that support the NOAA Level 1 requirements. These data are stored on-board the spacecraft as application packets in solid-state memory and transmitted to the ground as Stored Mission Data (SMD) via X-band (8 to 12 GHz) communications when the satellite is within range of a Control and Data Acquisition Station (CDAS), primarily at Svalbard Island, Norway with a backup site at Fairbanks, Alaska. The location of the Svalbard CDAS and the high-frequency downlink allows each orbit of SMD to be transmitted to the CDAS on each orbit (i.e., every 102 minutes). The system has sufficient additional capacity to store and download over two orbits of SMD in case a downlink is missed or SMD is lost in transit and must be retransmitted from the spacecraft.

The CDAS stores and forwards the application packets over commercial terrestrial communications circuits (primary and backup) to the Interface Data Processing Segment (IDPS), with IDP sites located at the NOAA Satellite Operations Facility (NSOF) in Suitland, Maryland,

the Air Force Weather Agency in Omaha, Nebraska, and the Ground Integrated Support Facility (GISF) in Aurora, Colorado. Each IDP independently generates Level 0 Raw Data Records (RDRs) by separating the application packets by type. The IDPs apply navigational information and calibration tables to the RDRs to generate Level 1 calibrated and geolocated Sensor Data Records (SDRs). The IDPs apply science algorithms and combine sensor data types to generate Level 2 Environmental Data Records (EDRs). The IDPs retain the data records for at least 24 hours.

The NDE DHS at the NSOF receives S-NPP data records based upon standing orders or subscriptions placed with the IDP at the NSOF. The IDP formats all data records in Hierarchical Data Format 5 (HDF5) for delivery. The IDP delivers the HDF5 data record files into designated file directories on the ESPC Storage Area Network (SAN). The IDP also delivers a checksum file for each data record file into the same directory, as well as Data Delivery Reports listing all files delivered in a configurable period.

Upon receipt of the HDF5 data, the NDE DHS produces and distributes all required products according to the production rules and subscriptions defined in its database pertaining to the AWIPS user profile. NDE's Data Selection Service (DSS) enables tailoring of products. Specific product tailoring that NDE provides to the AWIPS Community includes:

- Translation into a supported standard data format (e.g. HDF5 to NetCDF4);
- Geospatial sub-setting by pre-defined geographic regions (e.g. Alaska, CONUS, Caribbean/Puerto Rico, Pacific);
- Data aggregation and reduction (e.g. merging of sensor and geo-location data files into one file, elimination of unnecessary metadata, and modification to geo-location information); and
- Lossless data compression (e.g. Gzip) applied at the file level.

More NDE product specific criteria are outlined within Section 6 of this document. Once NDE product generation is complete, the system compresses all data files using the agreed upon methodology and, for AWIPS orders only, prepends the appropriate WMO Header (as defined by the AWIPS Data Manager) to each file. NDE delivers the NetCDF4 tailored data files into designated file directories on AWIPS FTPS servers. The NDE DHS also generates a Message Digest 5 (MD-5) checksum file after all tailoring steps are complete and delivers it into the same directory for each data file that was successfully delivered.

4.2 Data Subscription

The following steps outline the process for submitting and implementing a data subscription from the NDE DHS.

NEAR TERM process for submitting and implementing initial data sets:

1. The AWIPS Data Manager will coordinate data subscription requests with the NDE Data Distribution and Accounts Manager (DDAM). The DDAM will provide the AWIPS

Data Manager with a spreadsheet listing available products from the NDE DHS, and the AWIPS Data Manager will return the spreadsheet to the DDAM with requested products indicated per instructions with the spreadsheet. Alternatively, the AWIPS Data Manager may update a previously submitted request spreadsheet and deliver the update to the DDAM. The spreadsheet will constitute the data subscription request.

2. The DDAM will review the data subscription request with the ESPC Network Manager for approval and return questioned items to the AWIPS Data Manager for further discussion.
3. The DDAM will enter approved data request items into a subscription under the AWIPS Data Manager account on the NDE DHS and activate the subscription at a date and time coordinated with the AWIPS Data Manager and ESPC Network Manager.
4. The AWIPS Data Manager will verify and acknowledge first receipt of subscription data request items by e-mail to the DDAM. This will complete the data subscription transaction.

FUTURE:
The AWIPS Data Manager will request a data subscription from NDE DHS by submitting a Data Access Request Form to a designated ESPC representative, following instructions included on the form.

4.3 AWIPS Data Receipt and Handling

4.3.1 Data Dissemination

4.3.1.1 Data Dissemination Transaction Overview

4.3.1.1.1 Facilities

NDE is currently configured to use two FTPS clients, both of which may actively push S-NPP data products at a time. Each FTPS client is assigned a separate IP address. The NDE FTPS clients interface with the AWIPS NDE Ingest Service (ANDIS) at the NWS NCF facilities (ANCF, BNCF and TNCF). The ANDIS interfaces at the ANCF and BNCF are used for operational data transfer while the ANDIS interface at the TNCF is for testing and development only. ANDIS at each of the NCF facilities (ANCF, BNCF, and TNCF) is associated with a single "floating" IP address at each location. Each floating IP address is shared between the two NCF FTPS servers, only one of which is active at a time. Furthermore, ANDIS instances may be active concurrently, but NDE will only transmit operational S-NPP products to either the ANCF or the BNCF at any given time, but not both. The ANCF ANDIS instance will normally be used operationally. The BNCF instance will only be used during planned events, or during a failure that results in ANCF resources being inaccessible. NDE will switch between the ANCF and BNCF facilities only when directed to do so by NWS. This operation must be coordinated with the AWIPS NCF.

It should also be noted that NDE's only facility is at NSOF, within the ESPC, and there is no back-up site for NDE, but planning is in progress for a backup at Fairmont, WV.

4.3.1.1.2 AWIPS NDE Data Ingest Service

ANDIS is available on multiple servers within each NCF facility, but the ANDIS site IP address is only assigned to a single host at any given time. The Linux Heartbeat high availability cluster facility is used to manage the ANDIS service. Heartbeat monitors the health of cluster members (servers) and will automatically swap ANDIS to the backup server in the event that the active server becomes inoperable. Swapping ANDIS between cluster members is transparent to NDE. However, the detection of a server failure and the subsequent activation of the backup server are not instantaneous. In some instances, more than 60 seconds will be required to complete a server swap operation, during which time the ANDIS service will not be available to NDE.

4.3.1.2 FTPS Implementation Specifications

The FTPS protocol extension (RFC 4217) is used to transfer NDE products to AWIPS. NDE and AWIPS use the TLS cryptographic protocol. The data channel is not encrypted. The command or control channel is encrypted. The determined port ranges configured by the NDE and AWIPS servers to communicate and transfer data are specified in Table 4-1 below.

Interface	Parameter	Port Number/Range
ESPC(NDE)-ANCF/BNCF/TNCF	Protocol Specification	RFC 4217
	Control Channel/Connection	#21
	Data Channel	#40001-40500
	TLS Version	v1
	SSL Enabled?	Yes
	Mode	Passive

Table 4-1: FTPS Configuration Parameters

Each FTPS session is good for the delivery of only a single file and is closed after the file is delivered. The session is re-negotiated for each file or granule that needs to be sent. NDE is the client and ANDIS is the server for this interface. Authentication of the NDE client will be performed by operating system-level user name and password, which are provided over the encrypted FTPS control channel. Authorization to access files and directories by the NDE client on the AWIPS hosts will be granted by file system permissions on the AWIPS hosts. Access by NDE client systems to AWIPS systems will be limited to the identified hosts and output directories. A unique user ID and strong password, meeting applicable government requirements (https://access.portal.noaa.gov/doc/CIO/ITSITnew/CITR_009_Password_Requirements.pdf), will be established specifically for this interface.

The NDE and AWIPS software is designed in such a manner that multiple concurrent FTPS

sessions can be established by the NDE client to the AWIPS host and multiple files can be transmitted concurrently. NDE systems will transfer each file only once to a single operational AWIPS end-point (ANCF or BNCF). All files are written to a single AWIPS-specified directory location, which is a configurable parameter within NDE systems. Any changes to the destination directory must be coordinated between AWIPS and NDE staff. The AWIPS System Directory section is further described in section 4.3.1.3.

All product files contain a WMO header of the form "TTAAii CCCC DDHHMM<cr><cr><lf>", where <cr> is the ASCII Carriage Return character and <lf> is the ASCII Line Feed character. The WMO header is located at the beginning of each product file (file naming convention section). The WMO headers are further described in section 6 of this document. After each product file has been successfully written to the AWIPS host, a second file of the same name with an additional ".md5" extension is written to the same location on the AWIPS host by NDE. This second file contains an MD-5 hash of the product file of the same name. This process is described in detail in section 4.3.2 of this document.

AWIPS' ingest program introduces the product into the data stream that carries NDE products for uplinking over the SBN. All actions on product files (ingest, rejection, etc.) are logged with the time stamp and all appropriate information. Log files are archived for two weeks. Logs will be permanently deleted when they are more than 2 weeks old. Failure scenarios are described in detail in section 5 of this document.

4.3.1.3 AWIPS System Directories

The AWIPS ANCF and BNCF FTPS servers are configured with a standard directory for S-NPP products. This directory is currently set as /data_cs1/co/nde/ftp. NDE will place all S-NPP products into this directory at ANCF, BNCF and TNCF.

4.3.1.4 File Naming Conventions

The product naming convention is described in detail in section 6 of this document.

4.3.1.5 Data Dissemination Example

NDE is able to establish two parallel simultaneous sessions to any FTPS server for each S-NPP imagery file push transaction. The FTPS server is configured as passive/explicit. The ANDIS FTPS server listens on port 21 for control messages and TCP ports between 40001 and 40500 for data. The following data dissemination example outlines the steps associated with the successful transfer/push of a single S-NPP Imagery product file into the ANCF FTPS server at AWIPS. The process is identical for delivering S-NPP data products to either TNCF or BNCF.

1. The NDE FTPS client establishes a control session with the FTPS server, specified by the ANCF FTPS server IP and the destination port 21
2. The NDE FTPS client initiates the Transport Layer Security(TLS) authentication
3. The AWIPS FTPS server negotiates the TLS session, control encrypted and data not encrypted
4. The NDE FTPS client authenticates by providing the user credentials (username and password)
5. The NDE FTPS client establishes the "Passive" mode, which requires the AWIPS FTPS server to open another port for the client to connect to

6. AWIPS FTPS server opens a new TCP port from its range (40000-40500) and provides it to the FTPS client over the control session
7. FTPS client establishes a second session, the data session, using this TCP port provided by the FTPS server as a destination port to push the S-NPP data product to the FTPS server directory specified
8. Once the S-NPP file transfer is completed, the data session is closed
9. The control session is also closed after the data session is closed

After the delivery of the S-NPP product file to AWIPS FTPS server, the same process is repeated for the S-NPP product files associated MD-5 digest file.

4.3.1.6 Data Constraints

AWIPS NOAAPort is unable to process files larger than 200 million bytes (4,000 bytes/block * 50,000 blocks maximum).

4.3.1.7 Data Retention

NDE retains S-NPP data products and their MD-5 files for a period of up to 96 hours. After the 96th hour, these files are deleted.

AWIPS NCF receives two types of files from NDE; the actual sensor data file (which includes the geo location information) and the MD-5 file. The MD-5 files are deleted once the associated product file is processed. Data product files are stored using sequential file names that emulate a circular queue with a set number (*n*) retained. New products will overwrite older products with only the *n* newest products stored. The value of *n* is configurable, at this time in the 20-50 range (the AWIPS NCF has no archive requirement for the NPP/VIIRS imagery).

4.3.2 File Receipt Notification Transactions

4.3.2.1 Notification Transaction Overview

The NDE DHS notifies the AWIPS after DHS delivers a product file successfully. This is accomplished using the MD-5 file described earlier in subsection 4.3. After completion of a successful product file transfer from NDE DHS to AWIPS, the DHS pushes the corresponding MD-5 checksum file to the same AWIPS server and directory specified for delivery of the product file. The name of the MD-5 file is the same as the associated product file with the addition of the file type extension .md5.

The existence of the MD-5 hash file is the event signifying the availability of a product file. Product files are never to be processed if a matching MD-5 hash file is not received. ANDIS continually monitors the FTPS input directory for product files. Product files older than a defined and configurable age are relocated to a failure directory, with the action logged to the product log file for this interface. ANDIS monitors the NDE destination directory for the existence of hash files with ".md5" extensions. When AWIPS NCF detects a hash file, it moves its associated product file to another directory on the same file system that is polled by an ingest program and a message is logged to a standard AWIPS interface log file for tracking purposes. The hash file is then deleted.

Error correction mechanisms within the TCP protocol perform simple error checking during the

transfer. At this time NWS has no plans to use the MD-5 file to perform an integrity check on the received S-NPP product.

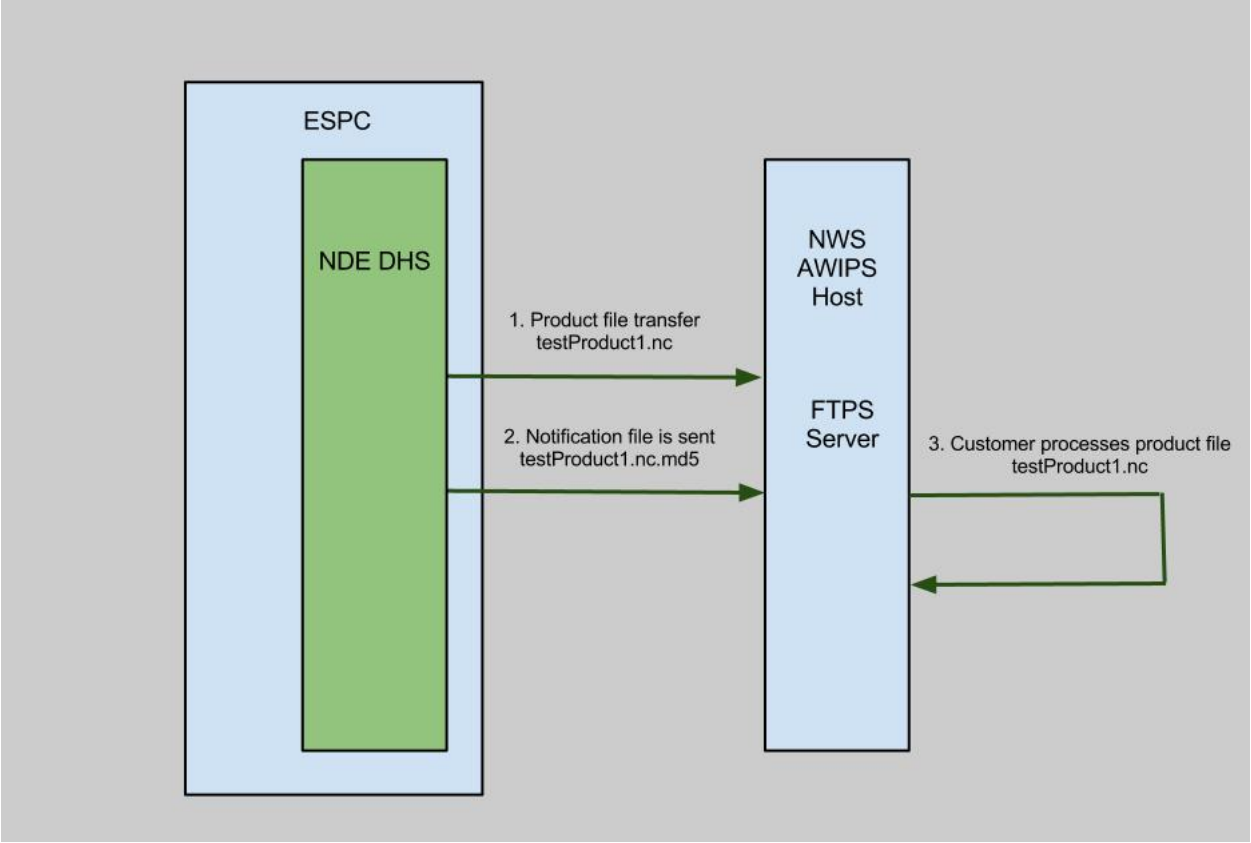


Figure 4-1: NDE DHS and AWIPS' File Notification Transaction Overview

4.3.2.2 Notification MD5 Naming Convention

The MD-5 notification filename is identical to the associated product filename <product filename> with the addition of the ".md5" file type suffix.

<product filename>.md5

4.3.2.3 Notification Example

Below are an example file and its associated MD-5 file.

Product File Name: <product filename>
Example:
VIIRS_I1_IMG_EDR_TIPB01_KNES_npp_s201206122253250_e201206122254590_c2012061
32011520.nc.gz.wmo

MD-5 File Name : <product filename>.md5

Example:
VIIRS_I1_IMG_EDR_TIPB01_KNES_npp_s201206122253250_e201206122254590_c2012061
32011520.nc.gz.wmo.md5

The content of the MD-5 file (hex): a1ee206fa4d0a9426e873d76214ae311

4.3.2.4 Notification Constraints

The MD-5 notification file contains a 128-bit hexadecimal value expressed as lowercase ASCII text in 16 octets (8-bit bytes). The value is generated by the Message Digest 5 hash algorithm described by RFC 1321 and the file is compatible with the MD-5sum utility included with many commercial and Open Source operating systems.

The notification file is only delivered after a successful transfer of the associated product file.

4.3.2.5 Notification File Retention

At AWIPS, MD-5 checksums are removed as soon as NCF has identified and found their associated product files.

At NDE the MD-5 hash files are kept for 96 hours and then removed.

4.4 System IP Address and Hostname Change

The system IP address and hostname change transactions are described in the “Interconnection Security Agreement Between NESDIS OSPO ESPC (NOAA5045) and NWS OST AWIPS” document.

4.5 CA Key Exchange Transactions

NDE and AWIPS each generate their own security keys. The key exchange between NDE and AWIPS is handled manually and out of band. The key generation and exchange between NDE and AWIPS is further described in the “Interconnection Security Agreement between NESDIS OSPO ESPC (NOAA5045) and NWS OST AWIPS” document.

5. Fault Resolution

Failure scenarios include authentication failure, authorization failure, connection failure, file transfer failure, and file system full failure.

5.1 Authentication Failure

Planned changes to authentication information must be coordinated between the AWIPS NCF and NDE since both systems must be updated.

Authentication failure will result in the NDE system setting their AWIPS interface “Down”. When NDE personnel are alerted to an authentication failure, they must contact the AWIPS NCF to coordinate a resolution.

5.2 Authorization Failure

An authorization failure indicates that the NDE FTP client does not have permission to access a requested resource on the AWIPS server. This usually means that the client does not have permission to access the FTP target directory. This can occur as the result of misconfiguration of the AWIPS host, or if a required remote file system is not mounted.

An authorization failure will cause the NDE system to set their AWIPS interface “Down” and requires manual intervention to correct. When NDE personnel are alerted to an authorization failure, they must contact the AWIPS NCF for resolution.

5.3 Connection Failure

Potential sources of fault transactions include loss of connectivity due to equipment failure or communications circuit failure, configuration error (i.e., human factors), and software failure (i.e., bugs). The potential impact is complete loss of service or corrupted/incomplete data transfer. The figure below depicts system procedures when transferring to the alternate host when a connection failure occurs.

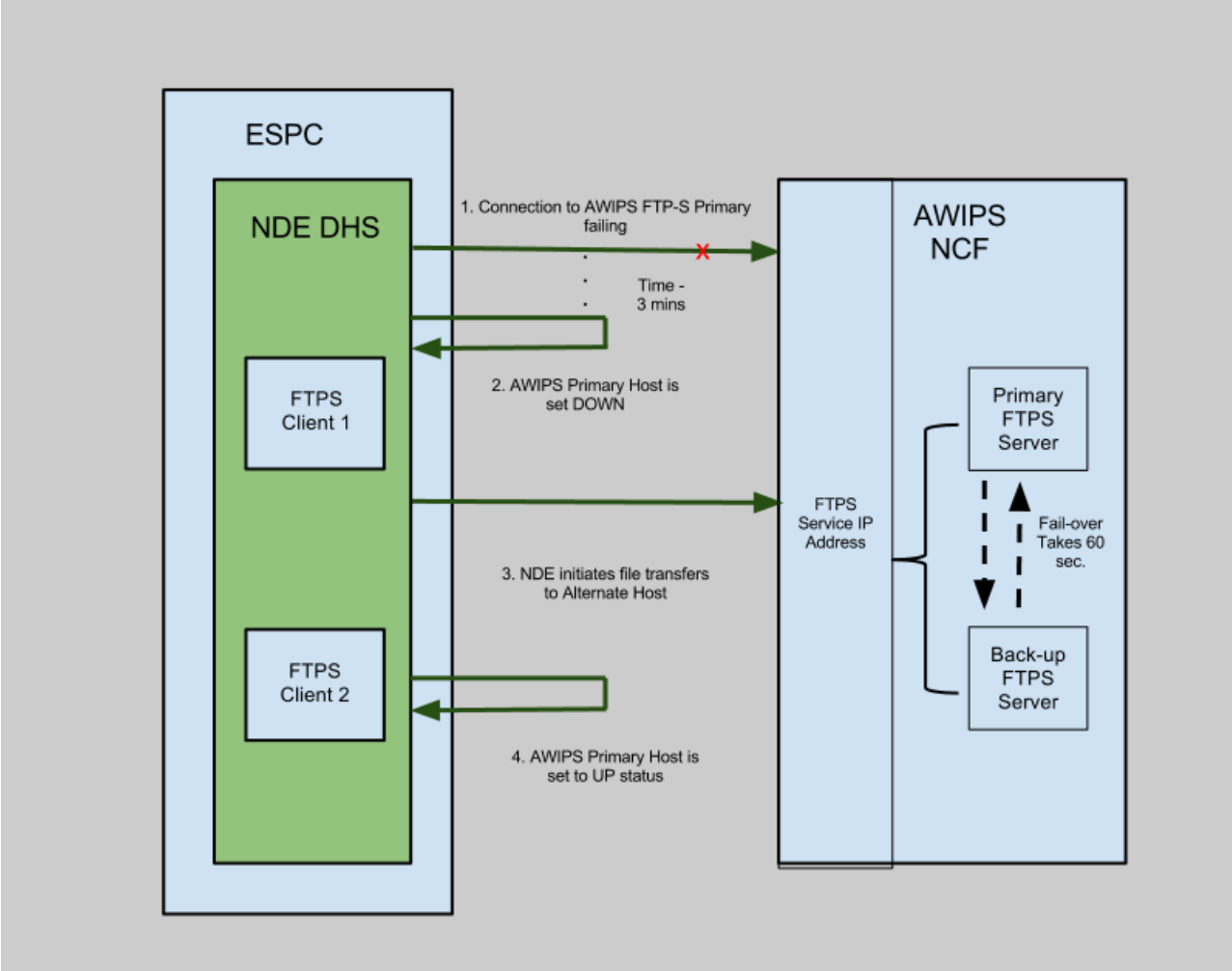


Figure 5-1: Connection Failure Scenario

The following describes the steps depicted in Figure 5-1: Connection Failure Scenario.

Note that for this interface, the AWIPS host designated as primary and alternate in the NDE database, have the same IP address. AWIPS failovers from one server to another are transparent in terms of its target IP. A service outage of up to 60 seconds may be experienced during an AWIPS system failover. The scenario depicted in Figure 5-1 will come into play any time a connection failure to AWIPS occurs and it may not indicate a failover condition.

1. The NDE FTP Client application detects a communications link failure to the AWIPS host designated as primary in the NDE database. NDE will attempt to connect to the primary for a maximum of 3 minutes before failing.
2. The NDE DHS Application sets the status of the AWIPS Host, designated in the NDE database as primary, to DOWN.
3. The NDE DHS Distribution servers stop attempting to Push files to the AWIPS Primary and instead use the AWIPS host, designated as alternate in the NDE database, which has status UP. If NDE has not designated an Alternate host for AWIPS in the NDE Database, NDE DHS will set the transfer in progress to FAILED and leave the remaining transfer jobs queued.

4. NDE operator notes that the AWIPS Host, designated as Primary in the NDE database, is DOWN and manually resets the status UP for operations.

5.4 File Transfer Failure

The following describes the steps depicted in Figure 5-2. Transfer fails to complete.

1. NDE DHS VSFTP Client begins a data transfer with some data successfully received on the target server.
2. If the transfer becomes idle for 15 minutes, the router may close the data or control connections causing a failure.
3. NDE DHS will set the transfer job status from RUNNING to QUEUED in order to retry the transfer again. Note that the target primary host is not set to DOWN.
4. The previously attempted transfer begins again to the AWIPS Primary.

The ESPC operations staff should be contacted if data problems are discovered and are associated with the ESPC data flow to AWIPS. A well-established call down list has been incorporated into ESPC operational procedures. ESPC Operations Help Desk can be reached at 301-817-3880, twenty-four hours a day. Change management for this interface will be coordinated through NOAA/NESDIS/OSPO/ESPC management. The NOAA/NESDIS/OSPO/ESPC contact is Russell Dyson, who can be reached at (301) 817-3901.

AWIPS support will be provided through Help Desk twenty-four hours a day and can be reached at (301) 713 9344. Change management will be coordinated through AWIPS CM group. The 8X5 (Monday through Friday) AWIPS contact is John Tatum. CM contact information is provided in Appendix B.

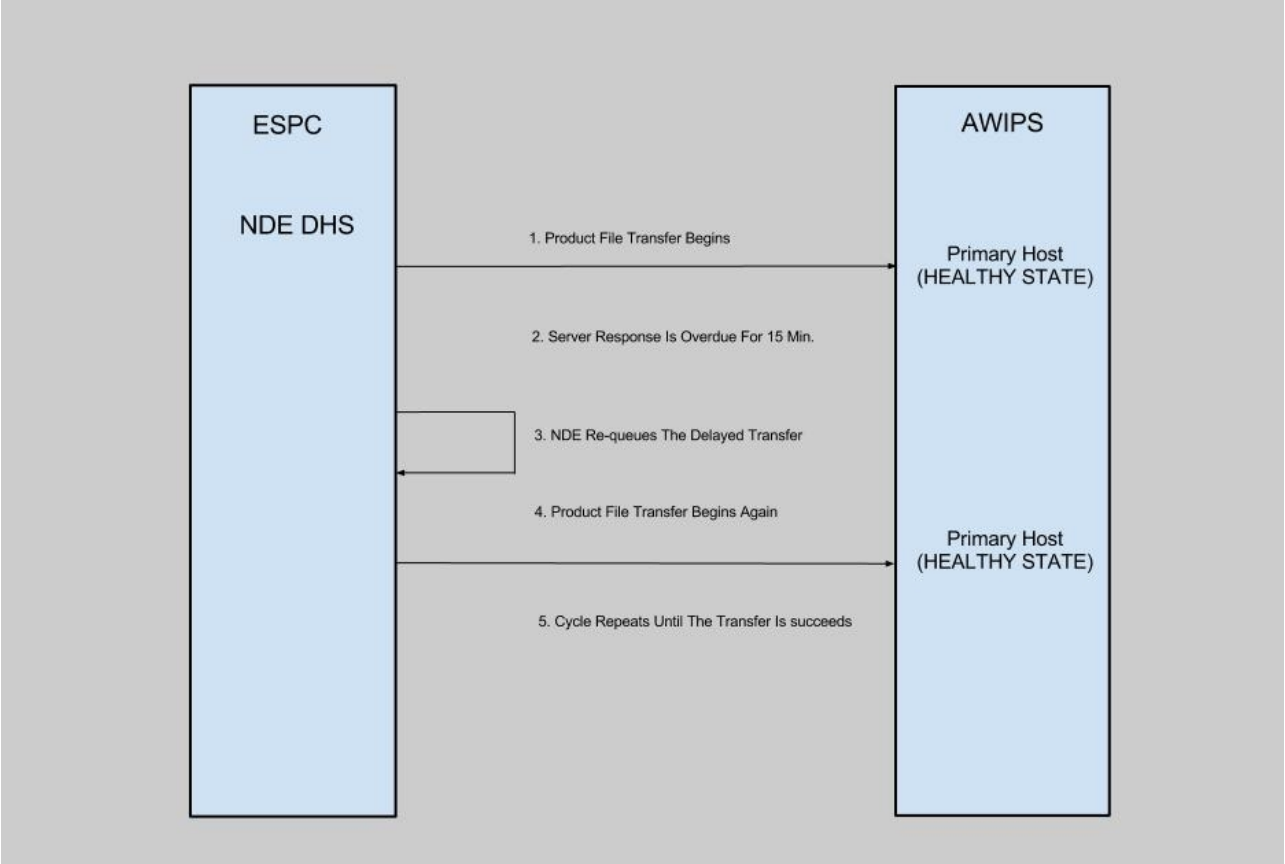


Figure 5-2: File Transfer Scenario

5.5 File System Full Failure

The NCF continually monitors the FTPS server input directories and file systems for capacity utilization. When these file systems reach 90% utilization, the Information Technology Officer (ITO) initiates an alarm. In the event of FTPS file system failure due the capacity overfill, the NDE FTPS push transaction will fail. The NDE will set the NDE-AWIPS interface status to "down", and an operator alert is generated. In this case, the NDE operator will need to contact the NCF operators to report and resolve the problem.

6. Tailored VIIRS Imagery Data Specifications

The Visible Infrared Imaging Radiometer Suite (VIIRS) instrument collects imagery and radiometric data for 16 moderate resolution channels, 5 high resolution imagery channels, and a near-constant contrast day-night band, for a total of 22 bands across the visible and infrared spectrum (see Table 6.1). There are 16 along-track detectors in the M-bands (moderate resolution channels) and 32 along track detectors in the I-bands (imagery or high resolution bands) resulting in spatial resolutions (at nadir) of 750 and 375 meters, respectively. The rotating telescope creates a ~3000 km x ~556 km (across-track x along-track) image of the Earth every ~86 seconds and this image is referred to as a VIIRS granule.

Band Number	**Wavelength -reflective (microns)**	**Band Number**	**Wavelength – emissive (microns)**
M1**	0.412	I4	3.74
M2	0.445	M12	3.70
M3	0.488	M13	4.05
M4**	0.555	M14**	8.55
I1	0.640	M15**	10.763
M5	0.672	I5	11.45
M6	0.746	M16**	12.013
I2	0.865		
M7	0.865		
DNB (Day-Night Band)	0.7		
M8	1.24		

M9**	1.378		
I3	1.61		
M10	1.61		
M11 To note: Band Numbers continue in column 3 of this table.	2.25		

*** - M-band channel has a corresponding GTM EDR product*
Table 6-1: VIIRS Channels

There are two varieties of VIIRS channel imagery. The standard Sensor Data Record (SDR) products and associated GEO product for both M-band and I-band channels contains all radiometric, geolocation, and quality information for each channel. The SDRs are used to create all the derived geophysical parameters known as Environmental Data Records (EDRs). The second variety of VIIRS channel imagery is the VIIRS Ground Track Mercator (GTM) EDR imagery products and associated GEO product. VIIRS GTM EDRs remap the VIIRS SDRs into a GTM grid of pixels where the rows are at right angles to the columns which are parallel to the S-NPP ground track. The pixel center spacing is constant at 375 meters for the fine GTM map (I-bands) and 750 meters for the course GTM map (M-bands). In addition, the "bow-tie" effect of the SDRs is not present in the EDRs (see NPOESS Common Data Format Control Book Vol. III for a description of the "bow-tie" effect). All high resolution I-band channels are remapped into GTM EDRs while only six (selectable) channels of moderate resolution M-band channels are converted into the GTM EDRs. The purpose of the GTM EDRs is to provide imagery for human viewing. Given the importance of cloud and surface imagery to forecasters, the VIIRS GTM EDRs are critical products for NWS field office use (e.g., AWIPS). The following sections provide details of the products, their attributes, and the NDE process for creating the VIIRS imagery required for AWIPS.

6.1 VIIRS GTM EDR Imagery Data Product

Given the large bandwidth required to distribute these products, NDE leverages its product tailoring capabilities to convert the HDF5 files into NetCDF4 files containing only the information that is required by AWIPS users and selects granules for distribution that intersect NWS-defined geographical regions.

NDE ingests VIIRS GTM EDR imagery from the IDPS in the form of HDF5 files. There is a file produced for every I-band and M-band GTM EDR channel for each granule (Earth image). An associated GEO file is produced for each resolution (M-band and I-band). NDE selects the granules that intersect the AWIPS regions (Table 6.3), opens the HDF5 files, reads the required

data and writes it to intermediate NC4 files. A tailored NC4 file is created for each required GTM EDR channel and the associated GEO information. Given the fixed resolution and gridded GTM layout of the EDR data, only three columns of geographical information (i.e., latitude and longitude) are required for properly mapping the EDRs. Selecting the first, middle, and last column of the GEO data and writing this to the NC4 file significantly reduces the bandwidth required for distribution. The full array of geographical information is rectified using these three columns by the AWIPS-II software. Reducing the amount of information enables combining the GEO data with the channel imagery into a single NC4 file for each channel. The combined file is then compressed using GNU Gzip and finally the appropriate 21 Byte WMO header is written to the compressed NC4 file.

6.1.1 Channel Requirements

A phased approach to introducing VIIRS imagery to AWIPS-II users has been adopted. The NWS has prioritized VIIRS GTM EDR imagery channels. The following table lists the AWIPS-II required VIIRS GTM EDR channels and their priority.

Priority	VIIRS Channels
1	I1, I4, I5
2	I2, I3, M9, M15, M16, DNB/NCC

Table 6-2: AWIPS VIIRS GTM EDR Imagery Channels

6.1.2 Regional Requirements

AWIPS users require regional VIIRS imagery information. The following table lists the NWS-defined geographical regions, in descending priority order. All incoming VIIRS granules that significantly intersect these regions will be delivered to AWIPS-II users.

Region	Latitude/Longitude (four corners)
Alaska	(45, 120), (75, 120), (75, -120), (45, -120)
Pacific	(-30, 120), (45, 120), (45, -120), (-30, -120)
CONUS	(25, -120), (55, -120), (55, -60), (25, -60)
Atlantic	(10, -120), (75, -120), (75, 10), (10, 10)

Region	Latitude/Longitude (four corners)
Caribbean/Puerto Rico	(0, -85), (30, -85), (30, -50), (0, -50)

Note: Latitudes and longitudes are expressed above in units of degrees. Longitude convention is such that eastern-hemisphere longitudes are positive and western-hemisphere longitudes are negative, with the prime meridian having a value of zero; latitude convention is such that southern-hemisphere latitudes are negative and northern-hemisphere latitudes are positive, with the equator having a value of zero**.**

Table 6-3: NWS Geographical Regions

6.1.3 NDE File Naming Convention

The following standard NDE naming convention is applied to all outgoing AWIPS-II files.

Field	Format
Product short name	I1: VIIRS_I1_IMG_EDR I2: VIIRS_I2_IMG_EDR I3: VIIRS_I3_IMG_EDR I4: VIIRS_I4_IMG_EDR I5: VIIRS_I5_IMG_EDR M9: VIIRS_M9_EDR M15: VIIRS_M15_EDR M16: VIIRS_M16_EDR DNB/NCC: VIIRS_NCC_EDR
WMO header	Varies by channel (see Section 6.1.4.3)
Data start date and time (UTC) of granule collection (satellite sense time).	sYYYYMMDDHHMMSSS
Platform ID	npp
Data end date and time (UTC) of granule collection (satellite sense time).	eYYYYMMDDHHMMSSS
NDE product file creation date and time (UTC)	cYYYYMMDDHHMMSSS

Extension(s)	.nc.gz.wmo

Table 6-4: NDE File Naming Convention for AWIPS-II VIIRS GTM EDR Products

For example, an NWS Alaska Region VIIRS channel I1 GTM EDR product file from June 6, 2012 at 22:49 UTC has the name:
VIIRS_I1_IMG_EDR_TIPB01_KNES_npp_s201206122249050_e201206122251470_c201207262003350.nc.gz.wmo

6.1.4 Tailoring Options

The following sections describe the specific tailoring NDE applies to all outgoing AWIPS-II VIIRS GTM EDR imagery products. The purpose of tailoring these products is to reduce the bandwidth required for distribution and to make the products simpler to work with. All tailoring components performed by NDE on VIIRS data sets destined for AWIPS were identified as product requirements by NWS.

6.1.4.1 AWIPS Defined VIIRS Variables & Attributes

NDE has the capability to read the incoming IDPS HDF5 VIIRS GTM EDR product files, select any variable and/or attribute and write them to the outgoing NC4 file. All variables have names in the form of *variable_name@[product short name]*. The following variables and attributes have been identified as required by AWIPS-II:

Channel Set	**Variable Names (data type): [Array Size]**	**Variable Attribute Names (data type)**
All	Latitude (float): [1541, 3] Longitude (float): [1541, 3] *For M-band product files the array is reduced to [771, 3]	long_name (string) missing_valuename (string) missing_value (string) units (string) valid_max (string) valid_min (string)
I1, I2, I3, DNB	Reflectance (ushort): [1541, 8241]	add_offset (float) scale_factor (float) long_name (string) missing_valuename (string) missing_value (string) units (string) valid_max (string) valid_min

		(string)
I1, I2, I3, DNB	ReflectanceFactors (float): [2]	long_name (string) units (string)
I4, I5	BrightnessTemperature (ushort): [1541, 8241]	add_offset (float) scale_factor (float) long_name (string) missing_valuename (string) missing_value (string) units (string) valid_max (string) valid_min (string)
I4, I5	BrightnessFactors (float): [2]	long_name (string) units (string)
M9, M15, M16	BrightnessTemperatureOrReflectance (ushort): [771, 4121]]	add_offset (float) scale_factor (float) long_name (string) missing_valuename (string) missing_value (string) units (string) valid_max (string) valid_min (string)
M9, M15, M16	BrightnessTemperatureOrReflectanceFactors (float): [2]	long_name (string) units (string)
DNB/NCC	Albedo (ushort): [771, 4121]	add_offset (float) scale_factor (float) long_name (string) missing_valuename (string) missing_value (string) units (string) valid_max (string)

		valid_min (string)
DNB/NCC	AlbedoFactors (float): [2]	long_name (string) units (string)

Table 6-5: AWIPS VIIRS GTM EDR imagery product file variables and attributes

6.1.4.2 Compression

NDE applies the standard GNU Gzip compression to every tailored NC4 file prior to attaching the 21 byte WMO header. GNU Gzip is a lossless compression algorithm based on the LZ77 and Huffman coding compression algorithms. More on Gzip compression can be found at www.gnu.org/software/gzip.

6.1.4.3 WMO Headers

NWS has assigned unique WMO headers for the three VIIRS channels (channels 1, 4 and 5) for the Alaska region. Additional WMO headers for other VIIRS channels and coverage regions will be assigned in the future. The current developed WMO headers are provided in table 6-6 below.

Geographical Region	VIIRS Imagery Channel	WMO Header
Alaska	Ch. 1 (0.64 um, Priority 1)	TIPB01 KNES
Alaska	Ch. 4 (3.74 um, Priority 1)	TIPB04 KNES
Alaska	Ch. 5 (11.45 um, Priority 1)	TIPB05 KNES

Table 6-6: Priority 1 WMO Headers For AWIPS (first 11 characters only)

6.2 NDE VIIRS Imagery Data Product Metadata

During the process of tailoring AWIPS-II VIIRS GTM EDR products, NDE writes identifying metadata information to the outgoing NC4 files prior to compression and the addition of the WMO header. The following sections describe the various metadata fields included in the global attributes of the NC4 file. The metadata is defined in XML format; a description follows.

6.2.1 Granule Level Identifying Metadata

Granule level metadata describes the dynamic identifying information for a granule produced by IDPS and tailored by NDE. The following attributes are included.

<attribute name="time_coverage_start" type=“string” value="Data start time in the YYYY-MM-DDThh:mm:ssZ format"/>
This attribute identifies the beginning date and time (UTC) the VIIRS sensor started collecting

information for the granule (first scan beginning time).

<attribute name="time_coverage_end" type=“string” value="Data end time in the YYYY-MM-DDThh:mm:ssZ format"/>
This attribute identifies the final date and time (UTC) data was collected by the sensor (last scan line end time).

<attribute name="date_created" type=“string” value="Date and time in the YYYY-MM-DDThh:mm:ssZ format"/>
This attribute identifies the date and time (UTC) the tailored product file was created by NDE.
<attribute name="start_orbit_number" type="int" value="1"/>
<attribute name="end_orbit_number" type="int" value="1"/>
These attributes identify the start and end orbit number, as defined by NPP IDPS, for the granule. Orbit numbers are incremented at the northward (ascending) equatorial crossing. It is possible for a granule to span two orbits.

<attribute name="day_night_data_flag" type=”string” value="Day, Night, Both"/>
This attribute identifies whether the data was collected during the day, at night, or during the transitory period between day and night.

<attribute name="ascend_descend_data_flag" type=”int” value="1 (descending), 0 (ascending)"/>
This attribute indicates whether data collection occurred during a polar ascending or descending orbit.

<attribute name="g_ring_latitude" type=”float” value="60.4, 59.1, 52.9, 50.8, 48.7, 54.2, 55.4, 57.8, 60.4"/>
<attribute name="g_ring_longitude" type=”float” value="-128.9, -156.5, -178.8, -176.3, -174.0, -153.6, -129.5, -129.3, -128.9"/>
These attributes define the latitude and longitude of a polygon that encompasses the geolocation of the granule. There are nine points where the first and last are the same in order to close the polygon.

<attribute name="id" type=”string” value="unique identifier for the granule (recommended UUID version 4)"/>
This attribute defines a unique identifier of the product and file. It is left blank at this time as there is adequate unique information for identifying NDE produced files.

6.2.2 General Information

The following static global attributes are included.

<attribute name="project" type=”string” value="NPP Data Exploitation"/>
This attribute describes the project that created the file. AWIPS-II NPP products are created by NDE.

<attribute name="title" type=”string” value="Name of product or product short name"/>
This attribute identifies the product short name, with the range of values defined Table 6.4.

<attribute name="summary" type=”string” value="Brief description of this product..."/>
This attribute describes the product in a sentence.

<attribute name="institution" type=”string” value="DOC/NOAA/NESDIS/NDE > NPP Data Exploitation, NESDIS, NOAA, U.S. Department of Commerce"/>
This attribute defines the institution responsible for creating the product file.

<attribute name="naming_authority" value="gov.noaa.nesdis.nde"/>
This attribute defines the naming authority.

<attribute name="satellite_name" type=”string” value="NPP"/>
This attribute defines the platform used to obtain the data.

<attribute name="instrument_name" type=”string” value="VIIRS"/>
This attribute defines the sensor used to collect the data.

<attribute name="creator_name" type=”string” value="DOC/NOAA/NESDIS/NDE NPP Data Exploitation, NESDIS, NOAA, U.S. Department of Commerce"/>
This attribute defines the organization that created the data file.

<attribute name="creator_email" type=”string” value="Project team member email(s)"/>
This attribute lists email addresses of the organization that created the data file. Questions regarding the product in terms of content or metadata should be addressed to this email address, the contact information of the developmental team.

<attribute name="creator_url" type=”string” value="http://projects.osd.noaa.gov/NDE"/>
This attribute is a URL for the organization responsible for creating the product.

<attribute name="publisher_name" type=”string” value=""DOC/NOAA/NESDIS/NDE NPP Data Exploitation, NESDIS, NOAA, U.S. Department of Commerce"/>
This attribute defines the organization that published the file.

<attribute name="publisher_email" type=”string” value="Project team member email(s)"/>
This attribute lists email addresses of the organization that published the data file.

<attribute name="publisher_url" type=”string” value="http://projects.osd.noaa.gov/NDE"/>
This attribute defines the organization responsible for publishing the file website.

<attribute name="Metadata_Link" type=”string” value="NDE product-specific output file name"/>
This attribute will be output product name as specified in the NDE-specific file naming convention (see section 6.1.3).

<attribute name="keywords" type=”string” value="NPP, NDE"/>
This attribute lists keywords used for search and discovery.

6.2.3 Processing Information

The following attributes provide information on the processing of the file.

<attribute name="history" type=”string” value="Created by NDE Data Handling System Data Selection Services"/>
This attribute defines the system used to produce the product file.

<attribute name="processing_level" value="NOAA Level 2 data"/>
This attribute defines the processing level of the product (level 1, 2, 3, etc.) as defined by NOAA. All tailored GTM EDRs are Level 2 products.

<attribute name="source" type="String" value=”FIRST_INPUT_FILE SECOND_INPUT_FILE THIRD_INPUT_FILE ETC”/>
This attribute identifies all products used to create the product file.

<attribute name="geo_group_name" type="String" value=”FIRST_INPUT_FILE SECOND_INPUT_FILE THIRD_INPUT_FILE ETC”/>
This attribute identifies the associated geo file for the product.
<attribute name="alg_version" type="String" value=”3.0.7”/>
This attribute identifies the build of NDE DHS DSS used to create this product file.

6.2.4 Climate and Forecast Conventions and Controlled Vocabularies Used

Climate and Forecast (CF) conventions are explicitly defined and maintained by Unidata (netCDF developers). The following attributes reference the conventions and data set discovery employed. This information is static.

<attribute name="Conventions" type=”string” value="CF-1.5"/>
NDE employs version 1.5 of the CF metadata conventions.

<attribute name="cdf_version" type=”string” value="4"/>
This attribute identifies the version of NetCDF used to create the file.

<attribute name="Metadata_Conventions" type=”string” value="CF-1.5, Unidata Dataset Discovery v1.0"/>
Conventions used will always version 1.5 of the CF metadata conventions and version and 1.0 of the Unidata Dataset Discovery.

<attribute name="standard_name_vocabulary" type=”string” value="CF Standard Name Table (version 17, 24 March 2011)"/>
Standard naming vocabulary is derived from the version 17 CF standard name table.

Appendix A: Acronym Definitions and Glossary

The following is a glossary of acronyms/abbreviations with descriptions as used in this document.

Acronyms Abbreviations	Terms and Definitions
AFWA	Air Force Weather Agency
ANDIS	AWIPS NDE Ingest Service
AR	Archived
ASCII	American Standard Code for Information Interchange
ATMS	Advanced Technology Microwave Sounder
AWIPS	Advanced Weather Interactive Processing System
AVHRR	Advanced Very High Resolution Radiometer
BUFR	Binary Universal Form for the Representation of meteorological data
CCR	Cloud Clear Radiance
CF	Climate and Forecast (Metadata Convention)
COARDS	Cooperative Ocean/Atmosphere Research Data Service
CM	Configuration Management
COPB	CLASS Operational Planning Board
CrIS	Cross-Track Infrared Sounder
DAP	Delivered Algorithm Package
DCN	Document Control Number
DDAM	Data Distribution and Accounts Manager
DHS	Data Handling System
DoD	Department of Defense
DOORS	Dynamic Object-Oriented Requirements System
DMSP	Defense Meteorological Satellite Program
DMZ	Demilitarized Zone (a subnet)
DSS	Data Selection Service

Acronyms Abbreviations	Terms and Definitions
EDR	Environmental Data Record
EOS	Earth Observing System
ESPC	Environmental Satellite Processing Center
ESPDS	Environmental Satellite Processing and Distribution Services
FGDC	Federal Geographic Data Committee
FNMOC	Fleet Numerical Meteorology and Oceanography Center
FTP	File Transfer Protocol
FTPS	FTP Secure
GigE	Gigabit Ethernet
GLB	Global
GVF	Green Vegetation Fraction
GVI	Global Vegetation Index
GOES	Geostationary Operational Environmental Satellites
HDF	Hierarchical Data Format
ICD	Interface Control Document
IDPS	Interface Data Processing Segment
IMG	Imaging
ISO	International Standards Organization
IT	Information Technology
ITO	Information Technology Officer
IV&V	Independent Verification and Validation
JPSS	Joint Polar Satellite System
LEO	Low Earth Orbiting
McIDAS	Man computer Interactive Data Access System
MD5	Message Digest Algorithm 5
MIRS	Microwave Integrated Retrieval System
MMC	Mission Management Center

Acronyms Abbreviations	Terms and Definitions
MODIS	Moderate-resolution Imaging Spectroradiometer
NAVOCEANO	Naval Oceanographic Office
NCDC	National Climatic Data Center
NCEP	National Centers for Environmental Prediction
NC	National Centers
NCF	Network Control Facility
NetCDF	Network Common Data Form
NH	Northern Hemisphere
NNDC	NOAA National Data Center (or in short, Data Center)
NPR	NOAA PRoduct
NSOF	NOAA Satellite Operations Facility
NUCAPS	NOAA Unique CrIS/ATMS Processing System
NUPs	NOAA Unique Products
NESDIS	NOAA's National Environmental Satellite, Data and Information Services
NPOESS	National Polar-orbiting Operational Environmental Satellite System
NPP	National Polar-orbiting Partnership
NWS	National Weather Service
NWSTG	NWS Telecommunication Gateway
NOAA	National Oceanic and Atmospheric Administration
OAIS-RM	Open Archival Information System-Reference Model
OSD	Office of Systems Development
OSDPD	Office of Satellite Data Processing and Distribution
PAL	Product Area Lead
PAMV	Polar Atmospheric Motion Vectors
PII	Personally Identifiable Information
POES	Polar-orbiting Operational Environmental Satellite
PoP	Point of Presence

Acronyms Abbreviations	Terms and Definitions
RFC	Request for Comments, formalized memoranda addressing Internet standards
RFC	River Forecast Centers
RSE	Remote Sensing Extension to FGDC
SA	Submission Agent
SBN	Satellite Broadcast Network
SA	Submission Agreement
SND	Sounding
SPSRB	Satellite Products and Services Review Board
SSMC	Silver Spring Metro Center
SST	Sea Surface Temperature
STAR	NOAA NESDIS Center for Satellite Applications and Research
TBA	To be assigned
TBD	To be determined
TOC	Telecommunications Operations Center
UCAR	University Corporation for Atmospheric Research
UTC	Coordinated Universal Time
VIIRS	Visible Infrared Imaging Radiometer Suite
WAN	Wide Area Network
WFO	Weather Forecast Office
WMO	World Meteorological Organization
xDRs	Any JPSS Data Record: Raw Data Record, Sensor Data Record, Environmental Data Record, Time/date/ephemeris Data Record, etc.
XML	Extensible Markup Language

Acronyms Abbreviations

Terms and Definitions

Appendix B: Change Management Contact List

The following personnel will represent NDE, ESPC and AWIPS as part of the change management process for the NDE-AWIPS ICD document. The information in this table is subject to change without any notice.

NDE-AWIPS ICD CM Contact List			
Organization	**Contact Name**	**Phone Number**	**Email address**
NDE	Hanan Jordan	301-817-3853	hanan.jordan@solers.com
ESPC	Dorothy Brown	301-817-3809	dorothy.brown@noaa.gov
AWIPS	John Tatum	301-713-1975 x 113	john.tatum@noaa.gov